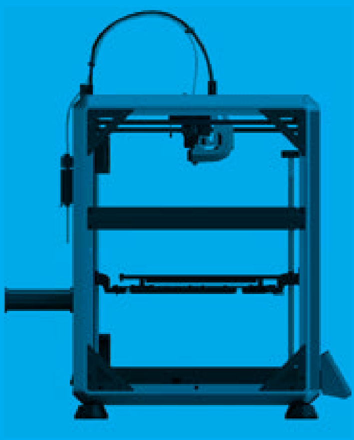
GIRO GROOVE DIY

Parts, *Materials* and Tools List.

See the complete tutorial at
girolimpo.com.br

Tools

You don't need a professional workshop to build the Giro Groove DIY. Some tools are essential, while others simply make the work easier — and some you can probably borrow, improvise, or easily find in your area.



3D PRINTER

Recommended print area: approximately 30 × 30 cm

The project includes parts that need to be produced using 3D printing. A printer with a print area of approximately 30 × 30 cm will meet the project's needs well.

Don't have a 3D printer?

Don't worry. You probably know someone who does.

That friend who bought a 3D printer and now prints everything that comes along can finally be useful. Or look for a 3D printing service in your city — you don't need to own a machine to build your Giro Groove.

LASER CUTTING

Some parts of the project are made from laser-cut MDF.

You don't need to buy a laser cutter — unless you're planning to start a business in this field.

Look for a laser cutting service in your city that works with MDF. Many companies accept digital files and deliver the parts already cut.

Tip: Search for “MDF laser cutting + your city name.”

WIRE CUTTERS

Used to cut wires and terminals and to make small finishing adjustments during assembly.

NEEDLE-NOSE PLIERS

Very useful for holding components, bending terminals, and reaching hard-to-access areas.

SOLDERING IRON

A simple 30 W or 40 W soldering iron is more than enough for this project.

There are only a few soldering points, and they are quite simple.

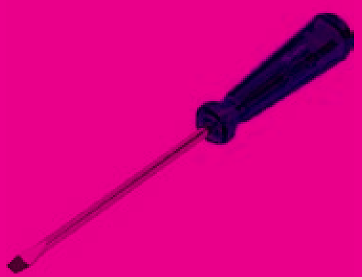
Never soldered before?

This could be a great opportunity to learn. But if you prefer, any electronics repair shop can do the job in just a few minutes. Since there are only a few connections, the cost is usually very low.

ELECTRONICS SOLDER

Use solder specifically intended for electronics.

It will only be used for the electrical connections in the project.



SCREWDRIVERS AND HEX KEYS

You will need the tools that match the screws you choose for the project.

They can be:

Flat-head screwdriver
Phillips screwdriver
Allen key

Or, if you prefer, a power screwdriver/drill with the corresponding bits. Besides speeding up the assembly, it also saves your arm some work.

RUBBER MALLET

Not an essential tool, but it can be very helpful for some fittings, allowing you to apply light impacts without damaging the 3D-printed parts or the MDF.

If you don't have one, you will probably be able to assemble the project just fine without it.

Electronic Components and Other Materials

3D Printing Filament and MDF

Let's start with the MDF. The parts must be cut exactly to the dimensions specified in the original files available on GitHub, with no changes to their size. The MDF must be 4 mm thick.

You can use coated MDF, which provides a better finish and eliminates the need for painting. If you choose regular MDF, that's fine too: after cutting, you will only need to apply paint or whatever finish you prefer.

The 3D printing filament can be virtually any type, but for this project we **recommend PETG because of its greater strength and durability**. The color is entirely up to you. Choose the one that best matches your Giro Groove.

Most of the electronic components and other materials are easy to find. You can look for them at electronics stores, Mercado Livre, AliExpress, eBay, or any other supplier you prefer.

The following list is fully illustrated, with photos and specifications for each component, specifically to make identification easier and help you avoid buying a different part than the one required for the project.

Make sure you get the correct model!

EG530SD-3F Turntable Motor

Flat drive belt for vinyl record turntables, 5 mm × 190 mm (measurement taken with the belt folded in half).

AT3600L Moving Magnet Cartridge with Stylus for Turntables, or any other compatible cartridge you may have available. The better the cartridge, the better the sound quality of your Giro Groove DIY.

DB15 female solder connector.

12V 1A 12W power supply with P4 output.

DC-022 female P4 power jack, 5.5 × 2.1 mm, threaded.

Mini DPDT toggle switch, 6-pin, ON-OFF-ON.

6202ZZ bearing, 15 × 35 × 11 mm.

608ZZ bearing, 8 × 22 × 7 mm.

625ZZ bearing, 5 × 16 × 5 mm.

2 metal male RCA plugs/connectors.

1 meter of Philips 2 × 0.50 mm² stereo audio cable.

60 cm of Philips 2 × 0.18 mm² stereo audio cable, or thinner.

25 machine screws with slotted, Allen, or Phillips heads. The head type does not matter, as long as you have the corresponding tool. What matters is that the screws are 15 mm long and, most importantly, have a 3 mm diameter.

2 M2.5 × 10 mm machine screws with M2.5 nuts for cartridge mounting. If you cannot find the screws, you can use double-sided tape to attach the cartridge to the headshell.

12 × 3.0 × 12 mm self-tapping screws.

1/4" × 1 1/4" hex bolt, black oxide, fully threaded, with 2 × 1/4" nuts.

1/4" × 5/8" hex bolt, black oxide.

M18 hex nut for making the counterweight.

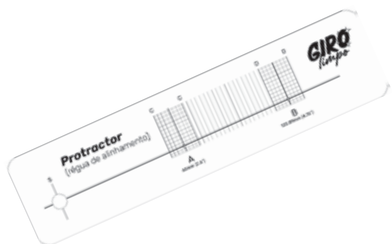
50 cm of automotive carpet or felt for
making the slipmat.

Super glue, epoxy, Super Bonder,
or another similar adhesive.

4 mm heat-shrink tubing.

And for the adjustments?

Stroboscopic disc for speed adjustment or
a smartphone app to adjust the platter RPM.



Protractor alignment gauge for cartridge alignment,
also available for download in the GitHub folder.

And now, it's time to get started.

With all the components and materials gathered, don't forget the 3D-printed parts. There are 20 printed parts in total that make up the project.

You will also need the 3 MDF pieces, already cut according to the original files.

Everything ready? Then it's time to get your hands dirty.

Visit girolimpo.com.br and head over to the Giro Limpo YouTube channel to follow the complete Giro Groove DIY tutorial.

And while you're there, take a moment to check out our store. If you'd like to support the project and help keep new projects like this coming, you can also contribute via PIX or PayPal.

And don't forget to like and follow the channel. It helps keep the project spinning.

GIRO limpo

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